

Vectorization in Therapy: Insights into Its Application in the Treatment of Huntington's Disease (December 2024)

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This work has been conducted with the support of M. Jean-Daniel Marty who accompanied us throughout the entirety of the project.

ABSTRACT Neurodegenerative disorders, including Huntington's Disease (HD), represent significant medical challenges due to their progressive nature and diversity of symptoms. Key challenges, such as crossing the biological barriers and the potential side effects, limit treatment options for these disorders. Huntington's Disease arises from expanded CAG repeats in the huntingtin gene (HTT), which leads to the production of a mutant protein, causing widespread neuronal damage and severe motor, cognitive, and psychiatric impairments. Current treatments are largely symptomatic and fail to alter the disease's course. The use of vectorization, particularly through adeno-associated viral vectors (AAV), is an innovative strategy of gene therapy for Huntington's Disease. Particularly, this article explores two studies where AAV5 vectors are employed to deliver therapeutic microRNAs that reduce the production of mutant HTT, mitigating toxic protein aggregation and neuronal dysfunction. Preclinical trials show that AAV5 is effective in lowering mutant HTT levels, enhancing cognitive and motor abilities, and increasing survival in mouse models. By enhancing targeted gene silencing and bioavailability, AAV5-based therapies pave the way for long-term solutions in HD and related disorders. Despite these advancements, significant challenges remain, including the immune response to viral vectors and the limitations imposed by vector payload capacity. This work emphasizes the potential of gene therapy through vectorization as a transformative tool in the treatment of Huntington's disease and similar neurodegenerative conditions.

INDEX TERMS Vectorization, Targeted Therapy, Huntington's Disease (HD), Neurodegenerative Disorder, Gene Therapy, Adeno-Associated Viral Vectors (AAV).

I. Introduction

NEURODEGENERATIVE diseases represent a group of complex pathologies characterized by the progressive degeneration of neurons, leading to irreversible impairment of motor, cognitive, and behavioral functions. Among these, Huntington's disease (HD) stands out due to its autosomal dominant genetic transmission and its devastating impact on the central nervous system. Described in 1872 by George Huntington, this disease is caused by a mutation in the huntingtin gene (HTT) located on chromosome 4. The mutation results in an abnormal expansion of CAG repeats, leading to the production of toxic protein accumulates in neurons, particularly in the striatum and cerebral cortex, ultimately causing neuronal death. [1]

The symptoms of Huntington's disease typically appear between the ages of 30 and 50, including involuntary movements (chorea), cognitive impairment (dementia), and psychiatric disorders (depression, anxiety). The disease progresses inexorably, leading to complete loss of autonomy and, ultimately, death within 10 to 20 years after diagnosis

[2]. Current treatments are primarily symptomatic and do not alter the disease's progression. In the context of advanced research, novel approaches such as vectorization have emerged as promising therapeutic strategies. Viral vectors, particularly adeno-associated viruses (AAV), are employed to deliver therapeutic agents directly into target cells. These vectors can cross the blood-brain barrier, making them especially suitable for treating diseases affecting the central nervous system. [1]

Current therapies aim to reduce the expression of the mutant HTT protein. One of the most promising techniques is RNA interference (RNAi), which utilizes microRNAs to target and degrade the messenger RNA of the protein. AAV vectors enable the delivery of these microRNAs, thereby reducing the production of mHTT and potentially alleviating symptoms. [3]

This prompts us to explore the biological mechanisms involved in these therapeutic approaches and how they could transform the treatment of neurodegenerative diseases.

II. Neurodegenerative diseases

Neurodegenerative diseases (ND) involve the progressive deterioration and loss of nerve cells, known as neurons. The term is made up of the words 'neuro', referring to neurons, and 'degeneration', which means the progressive deterioration in the structure or function of tissues or organs. The brain and nervous system, including the brain and spinal cord, are composed of neurons. Unlike many other cells in the body, neurons do not repair or replace themselves. Once destroyed or damaged, they are lost forever.

These diseases affect millions of people worldwide, particularly older adults, and have a significant impact on their well-being. Neurodegenerative diseases are often caused by complex factors, with no single cause identified. Specialists have identified various potential triggers and risk factors, grouped into categories such as ageing, genetics, environmental influences, lifestyle choices and personal medical history. ND are untreatable and disabling, as it leads to progressive neuronal loss or dysfunction. This can lead to problems with mobility (ataxia) or cognitive functioning (dementia). [4] [5]

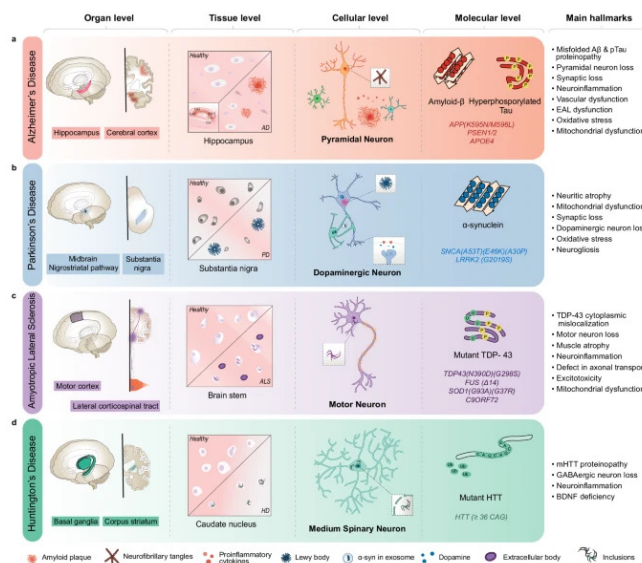


FIGURE 1. The similarities and differences in the main hallmarks of neurodegenerative diseases from the organ to molecular levels. **a.** Alzheimer's disease is marked by the buildup of misfolded amyloid- β and tangles in brain cells. **b.** Parkinson's disease is caused by clumps of misfolded alpha-synuclein and the loss of dopamine-producing neurons. **c.** Amyotrophic lateral sclerosis involves the buildup of mutant TDP-43 and other proteins, leading to the degeneration of motor neurons. **d.** Huntington's disease is caused by the mutant Huntingtin protein and leads to the degeneration of neurons. [6]

A. Examples of Neurodegenerative Diseases

There are many ND, each with specific symptoms and traits. **Figure 1** shows some of the main neurodegenerative diseases, but there are others, such as: Dementia with Lewy bodies (DLB), Corticobasal degeneration (CBD), Spinocerebellar ataxias (SCAs), Multiple system atrophy (MSA),

Progressive supranuclear palsy (PSP), among others. [5] [7] All these diseases shown different symptoms but most of them include motor symptoms, depression and anxiety and alterations of vital functions such as breathing and cognitive abilities.

Huntington's disease, the condition we will study in this research work, affects movement, emotions, and the ability to think. Huntington's disease (HD) is a rare neurodegenerative disorder with a global prevalence of approximately 2.7 per 100,000 individuals. [8]

B. Complexity of Treatments of ND

Now that an overview of ND has been established, it is essential to understand why ND are complicated to treat. Until now, there are no treatments to cure the diseases, but there are a few treatments for relieving symptoms.

1) Multiplicity of symptoms

As previously stated, ND are currently incurable. Nevertheless, some drugs are available to relieve symptoms. However, ND express themselves in many ways, which strongly complicates the evolution of medical research and pharmacology.

2) Molecular Interactions

Related to the multiplicity of symptoms, molecular interactions become a problem when the use of different treatments is involved. To illustrate, motor lesions, that are damaged regions affecting movement, are located in specific areas that may overlap with those corresponding to psychiatric manifestations or other kinds of symptoms.

In addition to interactions happening between two drugs prescribed for a ND, an important point to take into account is the molecular interactions between other types of medication, especially considering that ND develop with age. Taking medication also increases with aging, and the majority of patients are polymedicated, as shown by **figure 2**. For instance, a large part of the aged population takes beta-blockers to combat cardiovascular problems, but a review on biomarkers of Alzheimer's disease explains that beta-blockers are potentially a catalyst for cognitive decline as a function of the level of beta-amyloid deposits present in the brain [9] However, good maintenance of the cardiovascular system is as crucial as the maintenance of cognitive health. This is only one example showing the evidence of the major issue in taking medications acting on different areas of our body.

3) Side effects

Side effects are also a point that needs to be considered. According to the MSD Manuals, Vesicular Monoamine Transporter Type 2 (VMAT2) inhibitors are a treatment prescribed for Huntington disease that act on chorea symptoms, but they are strongly not recommended when patients are suffering from depression. [10] However, 39% to 53% of patients suffer from depression because of the disease. Other treatments exist and are proposed for these individuals;

however, their efficacy is often decreased. The impact of the VMAT2 medication results from the presence of regulatory neurotransmitters including serotonin, dopamine, and nora-drenaline [11], which are major contributors to depression and anxiety disorders. [12]

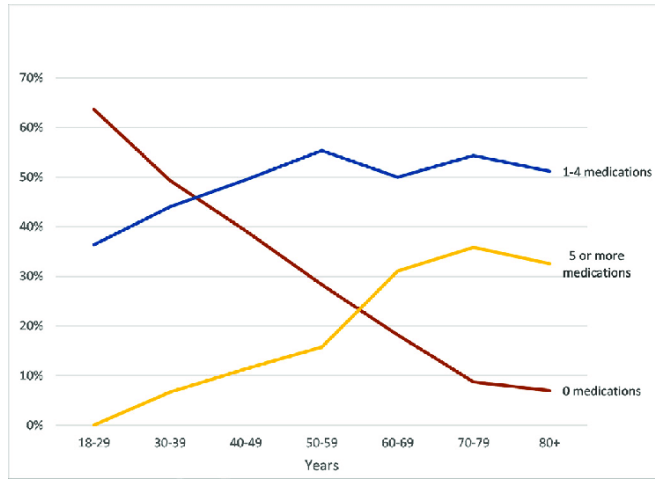


FIGURE 2. Relationship between age and medication number. Data are shown from 2014 to 2015. The graph indicates the relationship between a participant's age and number of medications taken. The x-axis indicates age and the y-axis indicates the proportion of the study population. The red line indicates participants taking 0 medications, the blue line indicates participants taking 1-4 medications and the yellow line indicates participants taking five or more medications. [13]

C. Biological barriers

Neurodegenerative biomarkers develop in the brain, leading to an overprotective environment due to multiple barriers present in our body that are responsible for the safety of our brain. **Figure 3** illustrates the composition of these barriers.

1) Blood-Cerebrospinal Fluid Barriers (BCSFB)

First, the Blood-Cerebrospinal Fluid Barrier (BCSFB) is composed of Choroid Plexus Epithelial Cells (CPECs), that form one of the interfaces between the blood and the nervous system, sealed by tight junctions that allow permeability. The choroid plexus is responsible for the secretion of cerebrospinal fluid, and its permeability is also strongly increased by the presence of proteins called claudin 1-2-3 [14]. Its role is to isolate blood from cerebrospinal fluid to avoid potential infection and contamination of the brain and to maintain the good quality of cerebrospinal fluid, which is responsible for the mechanical maintenance of brain integrity.

2) Blood-Brain Barrier (BBB)

The Blood-Brain Barrier (BBB) is currently the strongest barrier for therapeutic drugs to cross in order to reach therapeutic targets. It is mostly composed of brain capillary endothelial cells, that regulate exchanges between the blood circulation and surrounding tissues, astrocyte end-foot connections, that maintain brain vessels integrity, and other complex components. Capillaries are also sealed by

tight junctions that allow permeability, and this permeability is also strongly increased by the presence of claudin-5 proteins [14]. The role of the BBB is to maintain a constant environment in the brain, especially by maintaining homeostasis and preventing toxic substances from accessing the brain. The permeability and selectivity of the BBB are mainly responsible for the efficiency of the barrier but can unfortunately prevent beneficial drugs from reaching the brain. Nevertheless, several transportation mechanisms offer alternatives for nutriment delivery, and their understanding could allow for the development of drug delivery systems.

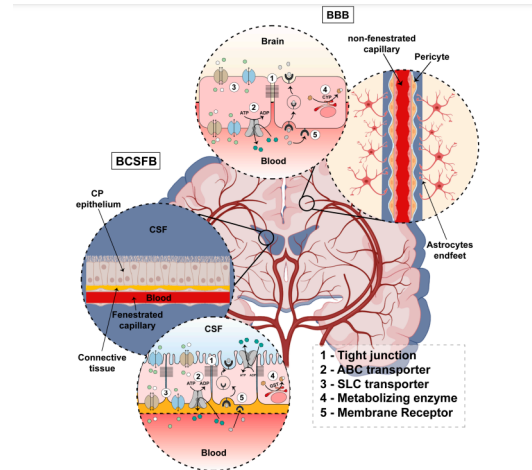


FIGURE 3. Schematic illustration of the blood–brain barrier (BBB) and blood–cerebrospinal fluid barrier (BCSFB). Both barriers are composed of tight junctions that control what can enter the brain and stop substances from slipping through the gaps between cells. Special transport proteins help move specific molecules across the barriers. Receptors help move substances by absorbing them or carrying them across cells. Lastly, enzymes in the brain can alter substances chemically to prevent damage. [15]

3) Transportation

To cross the brain barrier, several mechanisms can occur, as illustrated by **figure 4**.

-Passive diffusion: This process can be described as a natural and passive movement of particles from high concentration areas to low concentration areas without energy loss. However, it is limited by the hydrophobicity of elements, size, and specific environmental conditions.

-Active transporters: Active transporters use energy (often ATP) to bring elements from a low concentration area to a high concentration area. There are two major types of active transport in the BBB: ABC transporters (ATP) and SLC transporters (secondary transporters, no need for ATP). For example, P-glycoprotein (P-gp) is an ABC transporter which pumps drugs from the brain to release them into blood circulation [16].

-Mediated transcytosis transport: This process allows molecules, typically large or charged, to be transported across the barrier through specialized vesicular mechanisms. Receptor-mediated transcytosis involves a receptor on the cell's surface that binds to the substance, which is then

enclosed in a vesicle and transported across the cell, where it is released on the opposite side. Specific proteins can be transported by using receptors such as insulin receptors. Adsorptive-mediated transcytosis is triggered by electrostatic interactions and allows transportation of specific proteins and charged molecules by absorption process.

[17].

Multiple techniques have been tested to overcome the BBB barrier. However, it is a real challenge to pass through BBB without damaging it and putting the body in danger. BBB defects can lead to autoimmunity issues, edema and other disorders [16] but by using previous transporter's properties we can hope to find an efficient drug delivery system.

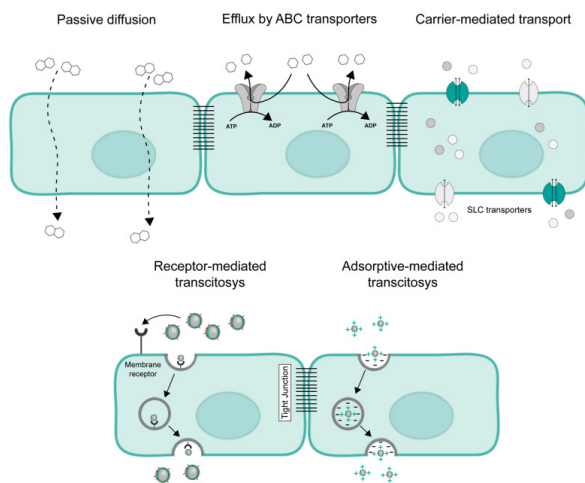


FIGURE 4. Mechanisms of transport across brain barriers. Small hydrophobic molecules can pass the brain barriers by passive diffusion. The ABC transporters use energy (ATP) to transport foreign substances, including therapeutic drugs. The solute carrier (SLC) transporters are responsible for carrier-mediated transport. Receptor-mediated transcytosis uses a receptor to bind to and transport the substance. Positively charged substances can cross the blood-brain barrier via adsorptive-mediated transcytosis. [15]

III. Huntington's Disease

Waters first described it in 1842, but it became known as Huntington's chorea in 1872 thanks to George Huntington. It is a hereditary disease that typically begins in adulthood and presents many symptoms, as shown in **figure 5**.

A link to chromosome 4 was identified in 1983, and the gene for Huntington's disease (HD) was found in 1993. This marked a major turning point in understanding HD and other neurogenetic disorders. For the first time, pre-symptomatic diagnosis became possible. As more diseases linked to CAG triplet repeats were discovered, HD became a model for many medical studies. The CAG triplet is a basic DNA sequence made up of cytosine (C), adenine (A), and guanine (G). [18]

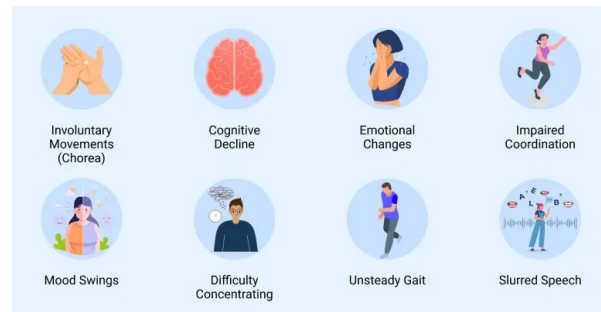


FIGURE 5. Common symptoms of Huntington's Disease [19]

A. Contraction of the disease

Huntington's disease (HD) is a genetic disorder passed from parent to child due to a mutation in a specific gene. If a parent has HD, each child has a 50% chance of inheriting the mutated copy of chromosome 4. The disease will not be passed on to children who do not inherit this mutation and they cannot transmit it to their own children.

When HD occurs without a family history, it is referred to as sporadic HD. Since the symptoms of HD typically appear in adulthood, individuals with expanded CAG repeats in the HTT gene may have the mutation without showing any signs of the disease. The hereditary transmission is shown in **figure 6**. [20]

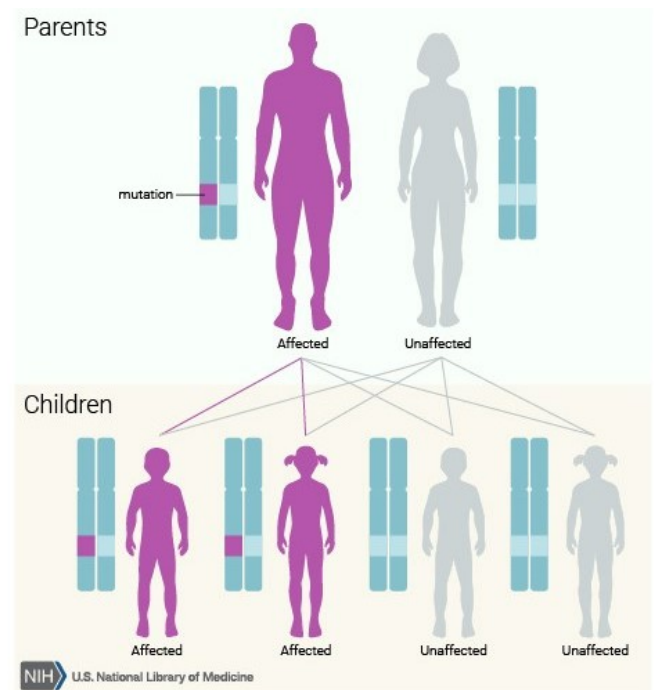


FIGURE 6. HTT gene inheritance. This condition is inherited in an autosomal dominant pattern, which means one copy of the altered gene in each cell is sufficient to cause the disorder. An affected person usually inherits the altered gene from one affected parent. In rare cases, an individual with Huntington's disease does not have a parent with the disorder. [21].

Huntington's disease is caused by a mutant gene which produces huntingtin. An abnormally high number of repeats of the cytosine-adenine-guanine (CAG) DNA sequence is caused by this mutation. (figure 7).

People who have fewer than 27 CAG repeats in their HTT gene are not at risk of developing HD. Generally, people with between 27 and 35 repeats do not develop the disease, but the CAG sequence may change. On chromosome 4 (4p16.3) is the HTT gene, which is responsible for huntingtin, a protein essential for the proper functioning of the brain, in particular for communication between neurons and postnatal development. Modification of the huntingtin protein can have harmful effects and disrupt its normal functions. When the huntingtin protein is mutated, it accumulates in clusters called inclusions, which form in different areas of the brain, usually in the nuclei of cells (intranuclear inclusions) and in their cytoplasm (intracytoplasmic inclusions). However, it has not yet been determined whether these inclusions play a direct role in the progression of the disease or whether they are simply a secondary effect of cell dysfunction. Another aspect of Huntington's disease is cerebral atrophy, which is a reduction in the size of the brain. This atrophy mainly affects the striatum, an important region for controlling movement. It is generally associated with a loss of neurons, which leads to the motor and cognitive disorders associated with this disease. In principle, when there are between 36 and 55 CAG repeats, the individual develops HD. However, those with more than 60 repeats often suffer from the juvenile form of the disease, which is an earlier form of Huntington's disease, meaning that symptoms can appear before the age of 20 or even as early as childhood or adolescence. This juvenile form of the disease is often more severe and progresses more rapidly. [20]

The age of onset of symptoms, particularly the first motor signs of huntington's disease, is inversely related to the length of the CAG repeats in the HTT gene. When the CAG repeats are longer, symptoms appear earlier and these repeats cause around 70% of the variations in age of onset. It is particularly in cases of paternal inheritance that the phenomenon of anticipation occurs, this phoneme being associated with the dysregulation of the CAG repeats during spermatogenesis. This dysregulation often causes the disease to appear in future generations at a younger age than that of the parent who passed on the mutant gene. There are three main categories of risk factors that impact on the onset of the disease: the length of the CAG repeats in the HTT gene, which is one of the most decisive causes, the instability of these repeats and the presence of genetic modifiers.

CAG repeats have a major impact on the course of the disease, particularly the development of cognitive, motor and neurological disorders, making them a key factor in the overall course of the disease. [8]

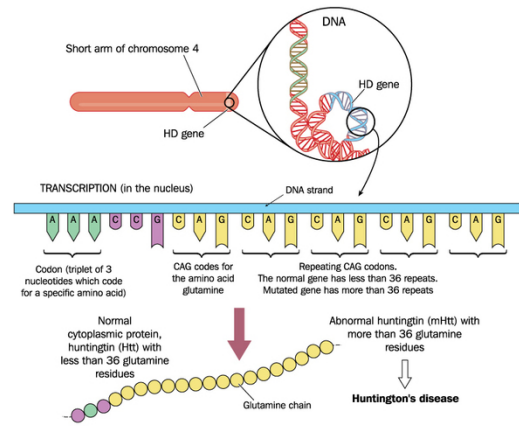


FIGURE 7. Gene mutation in Huntington's disease. The HTT variant that causes HD involves a DNA segment known as a CAG trinucleotide repeat, made up of a series of three DNA building blocks (cytosine, adenine, and guanine) that appear multiple times in a row. Normally, the CAG segment is repeated 10 to 35 times within the gene. In people with Huntington's disease, the CAG segment is repeated 36 to more than 120 times. [21].

B. Pathophysiology

Huntington's disease affects the neurons present in certain parts of the brain, such as the putamen, the caudate nucleus and the cerebral cortex. The disease manifests itself in involuntary movements called chorea, caused by the breakdown of specific nerve cells. Other types of nerve cells are lost, leading to problems such as dystonia (uncontrolled muscle movements) and akinesia (difficulty in moving). The symptoms of the disease vary from person to person, depending on how the nerve cells are lost. There seem to be several reasons for the onset of the disease at the same time. One key factor is the formation of neuronal aggregates, where abnormal clusters of mutant HTT protein accumulate inside the nuclei and cells, disrupting the ubiquitin-proteasome system, a cell-cleaning mechanism. Transcriptional dysregulation also plays a role, as errors in cellular development prevent the accurate copying and reading of genetic instructions. Another contributing factor is excitotoxicity, which results from the excessive release of glutamate, a chemical substance in the brain, leading to nerve cell damage. Mitochondrial dysfunction is also involved; when mitochondria no longer function properly, the energy required for nerve cells to operate correctly is compromised. Finally, axonal transport problems and synaptic dysfunction interfere with communication between nerve cells, preventing them from sending signals correctly to other cells. All these factors cause nerve cell damage in Huntington's disease. [8]

C. Specificity of HD

In the case of HD, all previous elements play roles in the complexity of treatment of the disease but more specific problems occur. The fact that HD affects all the peripheral systems and leads to inflammation and weight loss is not an issue present in all ND and makes this disease specifically hard to treat [22]. As explained, Huntington disease is due

to the repetition of a CAG repeat expansion in the huntingtin gene, that affects multiple cellular processes including transcription and the mitosis process which can lead to the expansion of the mHTT gene in somatic cells.

D. Therapeutic treatment in HD

Currently, the most common therapeutic approaches for HD are to relieve symptoms especially with drugs which target psychiatric disorders (antidepressants, anxiolytics...) and chorea (dopamine stabilizers). Physical reeducation is also often proposed for patients to help them handle their posture and balance. Concerning treatment to make HD regress, few research targets are pursued. Neuroprotection is an important research field which focuses on finding molecules to reduce death of neuron cells thanks to synaptic or neuronal transport. Cell therapy and neuron grafts try to implant neuronal cells to replace the ones that are destroyed by the disease. Finally, gene therapy is another approach led by scientists to try to find ways to reduce or suppress mutant genes by bringing a repaired one and it will be the one we will focus on [23].

E. Use of vectors for HD treatment

1) Why use a vector for HD?

As explained earlier, Huntington disease is characterized by a mutation of the Huntingtin gene. According to the Food and Drug Administration (FDA): "Human gene therapy seeks to modify or manipulate the expression of a gene or to alter the biological properties of living cells for therapeutic use." [24]. Actually, gene therapy includes a wide field of possibilities including inactivation, replacement or reprogramming of defective genes. Vectors are a perfect option for gene therapy because they can contain genetic code in their capsid and they have advantages such as precise targeting, easy administration, low immunogenicity, and long-term lasting effects. [25] First, gene therapy can be used to decrease levels of mHTT and avoid aggregates, especially by adopting RNA interference (RNAi) techniques. In this study led by Mireia Garriga-Canut et al. [26], reducing mHTT by using viral vectors showed good results. Synthetic zinc finger repressors (SFPs) are synthetic proteins created for preventing specific gene transcription. The major challenge of this approach is to target specific mHTT long repetitions without damaging normal genes. The use of a vector presents significant interest because of their precise targeting and their long action stability. The administration of ZFPs by vectorization showed a decrease of mHTT aggregates which contributed to reduced coordination symptoms and cognitive abilities decrease. Many other gene therapy approaches are experimented to treat HD. If reducing the level of mHTT by using RNAi seems to be a promising approach, clinical trials try to use vectors to maintain survival neurons and to increase neuronal creation to try to counterbalance neuronal loss linked to disease [27].

Aspect	Impact and Limitations
Nonuniform Transduction Patterns	Can limit vector efficiency and induce unequal therapeutic effects and bias treatment efficiency due to strong tropism from the vector.
Axonal Transport	Can lead to toxicity in other brain areas and induce neurodegeneration in unaffected structures.
Brain size and structure	Difficulty in reaching deep brain areas can limit transduction.
Immunogenicity	Can activate microglia, induce local inflammation, and accelerate the disease's progression.
Payload	Can lead to the use of truncated HTT, reducing efficiency and influencing disease progression.
Cross BBB	Requires modification and adaptation of the vector.

TABLE 1. Limitations of vectors and challenges in therapeutic approaches. [29]

2) Limits of the use of vectors for HD

Whereas vectorization for Huntington disease seems to be a perfect solution for treatment, a few biological challenges must be met, as illustrated in **table 1**. Adeno-Associated Virus (AAV) is the most common vector used in HD trials. Specific parameters of AAV need to be considered. Crossing the BBB is not easy for all vectors, AAV needs to be modified to be able to pass it. Payload is also a limiting parameter that can lead to the use of truncated HTT and reduce the efficiency. The following table is created based on results of two studies of AAV for HD [28] [26].

IV. Vectorization

A. Definition

Nowadays, it has become primordial to observe or treat at the smallest scale possible either to understand in a deeper way biological processes or to reach the main substance directly. In that way, vectorization has been developed to fulfill this need. Indeed, vectorization refers to the use of vectors to transport a molecule to a specific area or target in order to diagnose or treat a disease but also to observe particular biological patterns without activating any immune response. The first discoveries that led to vectorization as we know it today were made in 1928 by the bacteriologist Frederick Griffith through the "Griffith's Experiment" which provided the first hint that DNA could be the material of heredity. [31]

A vector is a nanoparticle of the size of a hundred of nanometers (2 nm - 100 nm) small enough to go through

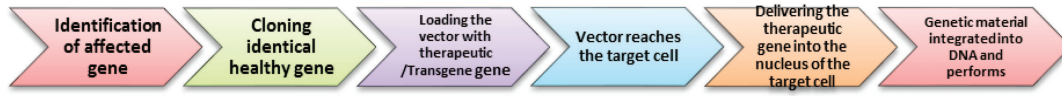


FIGURE 8. Schematic illustration of key steps in gene therapy [30].

blood capillaries. To do so, we use polymers, liposomes (small artificial spherical-vesicles formed by at least one lipid bilayer), aptamers (nucleic acids single-stranded (RNA or DNA), antibodies or even viruses as vehicles to transport our molecules. However, the vector's choice depends on multiple factors such as the purpose (diagnostic by imaging or therapy) and the aimed area.

Three categories of vectors are used depending on their abilities and features: first generation, second generation and third generation. The main issue when using vectorization is the immune response. Indeed, when diffused in the body, the first generation of vectors can be detected by the biological system because of the proteins's absorption located on their surface which indicates their presence. [32] Thus, their destruction is initiated and the targeting process can not be finalized. To counter this detection, the second generation of vectors has been developed. In that way, to avoid this phenomenon, we modify the nanoparticles by covering them with polyethylene glycol (PEG) polymers which will grant them a furtive character to dodge the absorption due to steric repulsion initiated by the polymer chains. [32] Nonetheless, the Enhanced Permeation and Retention effect (EPR) allows the vectors to diffuse towards the cell but also creates an overstacking around the cell due to the difficulty encountered by the vectors to regain the blood capillaries, mostly because of their width. As a result, we add "keys" (antibodies, sugar) on the surface of the vectors to allow recognition between vectors and thus, avoid overaccumulation. This new type of vector constitutes the third generation. [33]

B. Therapy

Vectorization for therapy has multiple applications but it is prominent in immunotherapy, chemotherapy and gene therapy. The main idea is to transport a substance to a pathogen cell. In cancer therapy, a tumor cell is hardly accessible by the active principle. Most of the time, the drug is eliminated even before reaching the pathogen cells and when the drug manages to get to them, it is expelled by the defensive mechanisms activated by tumor cells.

C. Gene Therapy

In this work, gene therapy will be the method used in the treatment of Huntington's disease. The process consists of "injecting naked DNA encoding the therapeutic protein" [34] to the sick cell. We can distinguish two types of gene therapy: gene transduction which uses viral vectors and gene transfection with the help of non-viral vectors. [35] The vector's role is to transport the therapeutic gene to the patient's target cell. Once it reaches the target cell, the

vector delivers the genetic material to the nucleus, where it integrates into the DNA and repairs the defective or mutated gene. Another way to proceed would be by preventing the cell from nourishing. The different steps in gene therapy's method are represented in **figure 8**.

We will now deal with the two different methods used for gene therapy: viral-mediated gene transfer and non-viral gene transfer.

1) Viral gene therapy

A virus is a 10 nm to 400 nm infectious agent composed of one or several nucleic acid molecules (DNA, RNA, single or double stranded) enveloped in a proteic capsule called "capsid". A nucleic acid is a chemical compound that can create, carry, encode or store the genetic information contained in the genome which constitute the genetic material by gathering chromosomes which also carry genes and thus the DNA or RNA. **Figure 9** illustrates the global gene configuration which gathers the mains compounds.

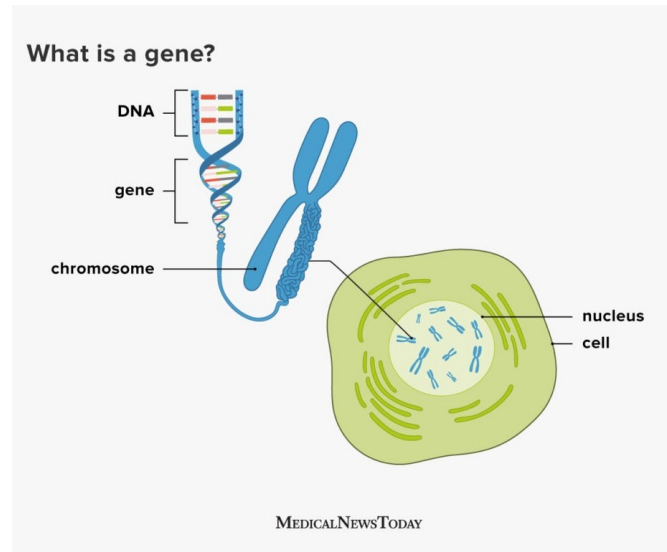


FIGURE 9. Schematic representation of a gene, illustrating the structural components, including coding sequences, and the genetic material necessary for protein synthesis. [36]

DNA (Deoxyribonucleic Acid) are double-stranded molecules that store genetic information of an organism by encoding instructions for protein synthesis and hereditary traits. It is located in the nucleus of eukaryotic cells (cell with a nucleus). Whereas, RNA (Ribonucleic Acid) are single-stranded molecules which act as a messenger (mRNA) to carry genetic instructions from DNA to the ribosome,

an intracellular particle that decodes genetic information transcribed by an mRNA to synthesis protein which ensure the cell's essential functions. [37] Both are nucleic acids.

Using viruses as a vector is a simple way to reach the cell target. Indeed, the main objective of a virus is to insinuate into a sane cell in order to replicate itself in quantity. The process occurring is such that the virus hangs on the cell, called the "host cell", allowing the nucleic acid to penetrate and liberate the virus's genetic material (DNA or RNA) which gathers the genetic code necessary for the replication. The virus has thus the ability to take control of the host cell by enabling the cell's function and forcing its own replication. In our case, we exploit the virus without the viral properties by deleting the viral genes (which allow the replication, assembling and infection) in order to replace them by the therapeutic ones. Nevertheless, by suppressing the virus's function, we suppress the replication which is problematic because vectors need to be produced in large amounts. This is why these vectors are engineered by using Packaging Cell Lines (PCL) whose purpose is to "replace a function of a deleted viral gene" [34] but also to ensure the vector's production excluding the production of the active therapeutic viral vector. To summarize, the final viral vectors are therefore obtained by deleting the viral genes and replacing them by the therapeutic genes from therapeutic molecules. Their replication has been inhibited, therefore vectors are inserted in packaging cells that have the ability to reproduce them. Finally, the vectors, produced in a large amount, are able to reach the pathogen cell and liberate the active principle after penetration [34].

2) Viral vectors

Viral vectors used for gene transduction (gene delivery exploiting viral vectors) are mostly adenoviruses, Adeno-Associated Viruses (AAV), retroviruses, lentiviruses and herpesviruses. The viral vectors used for gene therapy are represented in **Table 2**.

3) Non-viral gene therapy

Viral gene delivery is efficient but requires a lot of precaution when used because of the pathogenicity risk and insertional mutagenesis of the method whereas non-viral gene vectors have less immunotoxicity, their cost is cheaper and their production easier. Although non-viral vectors present less pathogenicity, the expression of their transgenes remains low which impacts the delivery's efficiency. [30] Nevertheless, advances are made in efficiency, safety and gene expression duration this treatment method.

The compounds grieved to a non-viral vectors are small DNA (Oligodeoxynucleotides), large DNA molecules (plasmid DNA), RNA (ribozymes, Si RNA, mRNA) [30]

4) Non-viral vectors

The main issue concerning vectorization, which also concerns viral gene therapy, is the crossing of the extracellular and intracellular barrier. Non-viral vectors must go through

these barriers in order to deliver the DNA or RNA to the nucleus and repair it. They are administered by transfection, microinjection or particle bombardment. The most common nucleic acid/vector combination used for non-viral gene therapy are DNA/Cationic lipid (Lipoplexes), DNA/Cationic polymer (Polyplexes) and DNA/Lipopolyplexes.

Cationic lipids are composed of hundreds of lipids, molecules which regroup a hydrophobic part and an hydrophile one, whose hydrophilic heads are positively charged. This polarization is the main feature that allows bindings with the nucleic acid which contain negatively charged phosphate groups. In that way, lipoplexes are created by the association of plasmids and liposomes. Lipoplexes (positively charged) interact with the glycoproteins and proteoglycans (negatively charged) present in membrane's cells which make the crossing easier. Moreover, lipids (positively charged) surround the therapeutic genes that have previously been inserted inside and provide them protection against nucleases, enzymes whose main purpose is to cut the phosphodiester link between two nucleic acid strands which causes the partial or total degradation of the nucleic acids.

We add PEG to balance the excess of charge present at the surface of lipoplexes which induces the reduction of the lipoplexes's half-life circulation in blood. As a result, we improve the delivery efficiency. Cationic polymers are positively charged groups that can interact with negatively charged nucleic acids to create nanosized complexes called polyplexes. Synthetic polymers, natural polymers and biodegradable polymers constitute the cationic polymers's type used in gene therapy to deliver DNA or RNA. [30]

Table 3 gathers some features related to lipoplexes (cationic lipids) and polyplexes (cationic polymers).

D. Immunotherapy

Immunotherapy for cancer works by sending triggering agents to stimulate the immune system and therefore to attack and annihilate tumor cells. The objective is to modify or reinforce the immune system. Unlike the following therapy methods that will be presented, vectorization in immunotherapy does not target directly pathogen cells. Cancer immunotherapy is intended to restore reactivity of the host's immune system to combat cancer.

Viral vectors (**table 2**) are widely used in cancer immunotherapy to trigger immune responses against tumors which results by delivering therapeutic genes to activate natural killer cells and T cells. Indeed, T cells are a crucial type of white blood cell (leukocyte) that play a central role in the immune response. Therefore, vectorization in immunotherapy consists to use viral vectors (mostly adenoviruses, retroviruses and lentiviruses) to deliver cytokine proteins, costimulation molecules and Tumour Associated Antigens (TAAs) to stimulate immune responses and thus to activate T cells whose purpose will be to neutralize tumor cells. [39]

Viral Vector	Characteristics	Advantages	Limitations	Applications
Adenoviruses	- Non-enveloped - Double-stranded DNA - Diameter: 70 nm - Genome: ~36 Kb - Complex and stable	- Efficient infection of human tissues - Widely used in gene therapy - Especially effective for respiratory diseases	- Short-lived due to early immune response - Limited transduction - Lack of long-term expression in dividing cells - Disappear within 24 hours after injection	- Cancer (pancreatic, breast, glioma, ...) - Respiratory diseases
Adeno-associated viruses (AAV)	- Non-enveloped - Small virion - Diameter: 20 nm - Genome: ~4.5 Kb - Construction: insertion of the therapeutic gene which replaces the coding sequences	- Remain passive without helper virus - Easier to package therapeutic genes - Low immune response risk - Long-term expression - Simpler manipulation due to the genome size	- Limited genome size - Not suitable for large genes	- Huntington - Rett Syndrome
Retroviruses	- Derived from RNA viruses - Construction : replacing replication genes with therapeutic ones - Non-replicating vectors - Integrated into host cells	- Stable therapeutic gene delivery - Suitable for inherited and chronic diseases - Good encoding expression	- Risk of mutagenesis - Potential over-expression issues	- Inherited diseases - Chronic diseases - Cancer
Lentiviruses	- Engineered to improve retroviral limitations - Can infect non-dividing cells - Large genetic payloads - Effective host genome integration	- Stable and long-term transgene expression - Non-inflammatory gene transfer	- Potential safety concerns due to genome integration	- Parkinson - Alzheimer - Polycystic Kidney Disease (PKD)
Herpesviruses (HSV)	- Enveloped double-stranded DNA - Genome size: ~150 Kb - Replicates in epithelial cells - Construction : cloning the therapeutic gene into a plasmid which contains the HSV - Helper virus HSV for transfection	- Large packaging capacity - Long-term transgene expression - Low toxicity due to latent state	- Enable the genes of interest by entering in a latent state - Latency can cause activation issues	- Painful Diabetic Neuropathy (PND) - Sensory Neuropathies (SN) - Cancer

TABLE 2. Comparison of Viral Vectors: characteristics, advantages, limitations and applications [34] [35] [38]

Type of Vector	Key Mechanism	Target Tissue	Advantage	Limitations
Lipoplexes (Lipid-based)	- Endocytosis - DNA condensation	- Airway epithelial cells - Endothelial cells - Hepatocytes - Muscle cells	- Safety - Low cytotoxicity	- Low to medium efficiency - Immunogenicity
Polypeplexes and Dendrimers	- Endocytosis - DNA condensation - Protein sponge effect	- Lung - Oral cavity	- Low immunogenicity - Fair efficiency	- Complement activation - Low efficiency - Cytotoxicity

TABLE 3. Comparison of Non-Viral Vectors: Mechanisms, Target Tissues, Advantages, and Limitation

Cytokines are synthesized or cell-secreted proteins which act as messengers between cells of the immune system while costimulation are cell surface proteins that are essential by providing the second signal of the two signals required to T cells activation. [39] TAAs are proteins found on cancer cells which, when detected, will generate the production of antibodies by the immune system. In our case, the viral vector injected will carry a molecule targeting it. Then, the genetic material will be transferred into immune cells like T cells or dendritic cells (“an immune cell that boosts immune response by showing antigens on its surface” [40]). Once inside, immune cells produce TAAs and display them on their surface to train the immune system to recognize them and thus, to attack cancer cells expressing the same TAAs.

E. Chemotherapy

For the following, the information has been taken from the study conducted by S. Collins and colleagues, “Viral Vectors in Cancer Immunotherapy: Which Vector for Which Strategy?” (2008), published in *Current gene therapy*. [39]

In chemotherapy, unlike immunotherapy, we use vectors to directly target tumor cells by delivering an active principle whose purpose will be to annihilate tumor cells. To do so, the use of aptamers as therapeutic agents has been developed to treat prostate, breast, lung and cervical cancer. Aptamers are nucleic acids whose specificity is their recognition and high capacity to bind to targets by folding into a particular structure. Aptamers behave like antibodies by recognizing molecules and cells. Therefore, aptamers provide an excellent platform to deliver the active principle by . As a result, the vector composition can be configured either in an aptamer-drug system or an aptamer-nanomaterial system.

Doxorubicine (DOX) is the most used anticancer drug in chemotherapy. Indeed, it has the ability to disrupt the transcription and replication by intercalating into tumor cells DNA. In that way, the aptamer-DOX combination by a non-covalent conjugation possesses high payloads and can reduce side effects on non-target cells. Nonetheless, this non-covalent conjugation brings some limitations like the unpredictable drug released by the aptamers when approaching the targeted cells. To hurdle, covalent bonds are used by exploiting ester, amide or disulfide. Nevertheless, the payload capacity becomes less relevant which results by the decrease of the targeting efficiency. This is why nanomaterials are employed to surpass these limitations. Aptamers-nanomaterials systems improve delivery efficiency by getting through the EPR effect (for nanomaterials with a certain range of diameters) and by grafting them with aptamers whose surfaces have been modified. By nanomaterials, silica nanoparticles, liposomes and polymer micelles are the ones being combined and conjugated inside of the nanoparticles by electrostatic attraction or covalent conjugation.

F. Diagnostic

If vectors are often used for therapy, diagnosis also presents an interesting application for them. In medical imaging, vectors are used especially for fluorescence microscopy (especially in the biology field) but also for nuclear medicine. Indeed, radioelements need to be carried by vectors to cross the cellular membrane, but they are also used to specifically target the organ that must be explored. The radio-isotope is encapsulated in the vector and injected into the patient's body to perform functional imaging.

G. Choice of vectors

The choice of vector is obviously limited by the size of the vector, biocompatibility with the body (especially to avoid risks of toxicity and allergy), and compatibility between the vector and radio-isotope. Vectors for imaging are mainly composed of colloidal sulfur, rhenium, or human albumin. Ideal fixation between vectors and radio-isotopes is assured by the presence of covalent bonds [41].

H. Functional imaging

Functional imaging includes PET scans, SPECT scans, or CZT cameras. These methods are emission imaging techniques that require the introduction of a radio-isotope into the body to be fixed on a specific organ and detected by a radioactive receptor camera. For these techniques, not every exam is done with vectors because some radio-isotopes can be administered orally or by inhalation. However, vector use is much more specific and efficient when there is a need to target a specific area of the body.

V. Analysis of a Therapeutic Vector for Huntington's Disease

Removing the mutant HTT expression is presently recognized as essential for treating HD. The monogenic characteristic of HD facilitates the development of therapies that modify the disease, targeting the elimination or reduction of HTT production. Therapeutic approaches utilizing RNA interference (RNAi) allow gene silencing after transcription has occurred, which would reduce the translation of toxic mutant HTT proteins. For this purpose, viral vectors are often employed as delivery agents of the RNAi products, to allow targeted administration and stability of the genetic package. [1]

To illustrate the advancements in this field, we have selected the ongoing research conducted by Melvin M. Evers and colleagues at uniQure biopharma B.V., Amsterdam, the Netherlands, in collaboration with Leiden University Medical Center and Lausanne University Hospital (Miniarikova et al., 2017 [1]; Spronck et al., 2019 [3]). The combined studies explore the use of adeno-associated virus serotype 5 (AAV5) as a vector to deliver interfering microRNA designed to target the mRNA of the mutant HTT protein in Huntington's disease mouse models. The structure of the

vector and the RNA interference sequence are crucial for treatment effectiveness.

The mitigation of HD-like symptoms and improved survival seen in HD rodents administered AAV5-miHTT reinforce the promising role of vectorization in the treatment of neurodegenerative disorders. [3]

A. Overview of the Therapeutic Vector

Adeno-associated virus serotype 5 (AAV5) is widely used in gene therapy due to its safety profile and efficiency in delivering genetic material into human cells. In the context of Huntington's disease, AAV5 has been engineered to carry therapeutic microRNAs (miRNAs) that specifically target the mutant Huntingtin (HTT) gene, responsible for the production of a toxic protein causing neuronal damage. Research conducted in rodents and non-human primates has demonstrated that AAV5 can effectively transduce the brain, positioning it as a promising option for RNAi-based gene delivery. [1].

B. Characteristics

The AAV5 vector is derived from a naturally occurring adeno-associated virus, but it has been genetically modified to eliminate its replication capability, making it safe for therapeutic use. Structurally, AAV5 has an icosahedral shape (polyhedron with 20 faces) with a diameter of approximately 20-25 nanometers, illustrated in **figure10**. [35] The small size and spherical symmetry allow it to penetrate deep into neural tissue, facilitating efficient delivery of its genetic cargo. This vector can encapsulate up to 4.7 kilobases of DNA, which is sufficient to carry therapeutic genes or regulatory RNA sequences designed to silence the expression of harmful proteins [3].

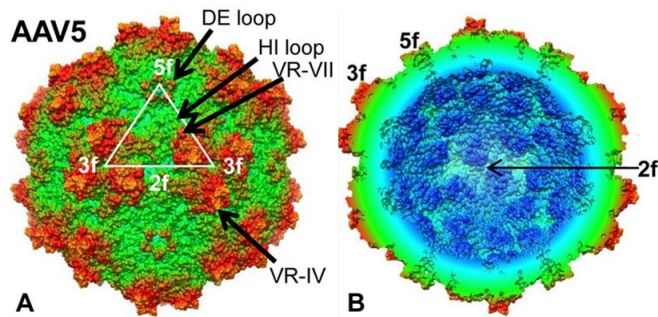


FIGURE 10. 3D representation of the Adeno-Associated Virus serotype 5 (AAV5). A. Exterior capsid surface for, radially color cued (from capsid center to surface: blue to green to red to yellow). B. Cross-sections of the AAV5, showing its interior surface. [42]

C. Manufacturing Process and Optimization

The production of AAV5 vectors involves a process of co-transfection in host cells, typically HEK293 cells. This method includes introducing the vector genome and helper plasmids into the cells to generate high-titer viral particles.

The process is then optimized through several purification steps, including ultracentrifugation and chromatography, to isolate pure AAV5 particles while minimizing contaminants [43]. Advances in vector design and production have improved yield, purity, and safety, making AAV5 one of the most reliable vectors for gene therapy applications in neurodegenerative diseases.

AAV5 vectors expressing the different miRNAs and GFP were produced by baculovirus-based AAV production system. The genetic material for the miRNAs was inserted into a transfer plasmid to create an entry plasmid, which serves as vehicle. These expression cassettes were then added to a baculovirus vector by homologous recombination, and the best clones were selected through plaque purification. The recombinant baculovirus was subsequently cultivated in significant quantities and the clones were tested for stability. To generate AAV5, the cells were infected with different baculoviruses carrying the necessary genetic material, including the vector genome, the enzyme for replication, and the protein needed to form the virus shell. After infection, the cells were broken open and treated with Benzonase to break down unwanted material. The AAV5 virus was then purified using a column and purification system. The amount of virus produced was measured using quantitative polymerase chain reaction (qPCR), a technique for measuring the amount of DNA. [1]

In later stages, the gene expression in the AAV5 vector was driven by chicken β -actin promoter, which ensures high levels of gene expression, and cytomegalovirus enhancer, which helps activate the gene. The transcription unit was surrounded by two AAV-derived inverted terminal repeats (ITRs), special sequences that assist the virus in packaging and delivering the gene into cells. [3]

D. Vector Content: Therapeutic Payload

To generate the therapeutic payload of the AAV5 vector in Huntington's disease, two approaches were considered: allele-selective knockdown and total-knockdown.

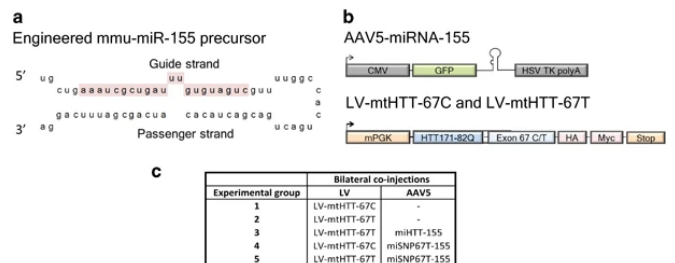


FIGURE 11. Design of a proof-of-concept study using the AAV5-miHTT-155 and AAV5-miSNP67T-155 vectors in an acute HD rat model. a. The structure and sequence of the engineered mmu-pre-miR-155 precursor. b. Schematic representation of the AAV5-miHTT-155 and AAV5-miSNP67T-155 expression cassettes; and LV-miHTT-67C and LV-miHTT-67T expressing the mutant HTT sequences. c. Experimental groups for bilateral co-injections in the striatum of rats. [1]

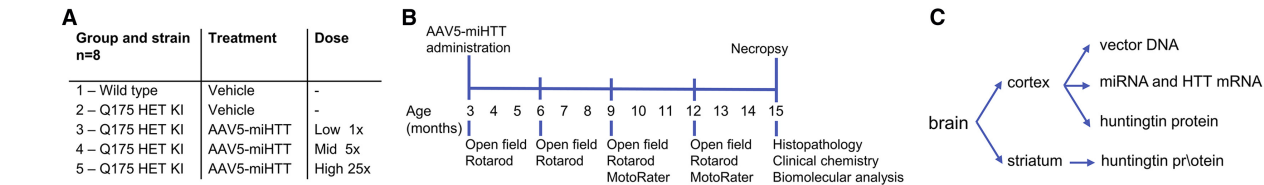


FIGURE 12. AAV5-miHTT Administration in the Mouse Brain Using a Single Bilateral Intrastratial Stereotactic Injection. **A.** Experimental setup showing wild-type and Q175 KI mice used in the study with the different dose groups. **B.** Experimental setup. Behavioral tests were performed to assess motor skills at several time points as indicated. **(C)** Brain sampling overview. The striatum and cortex were dissected from the rest of the brain. [3]

Allele-selective knockdown targets the mutant expression of the HTT gene and stops the production of toxic HTT protein while sparing the expression of the normal allele, which is essential for normal cellular function. Allele-selective knockdown uses single nucleotide polymorphisms (SNPs) that are unique to or more frequently found in the mutant allele to target it. [1].

Total-knockdown reduces or eliminates the expression of both normal and mutant alleles of the HTT gene. RNA interference (RNAi) is applied, where a micro-RNA, referred to as miHTT, binds to the mRNA of the HTT gene to silencing gene expression after transcription has occurred. Total-knockdown decreases the overall levels of HTT protein, which has therapeutic interest. [3].

Several therapeutic miRNAs were previously tested for their efficacy in HTT-knockdown in vitro, and those with strongest performance were selected for the subsequent studies in mouse models. The selected constructs were designed in the engineered mmu-miR-155 precursor. Namely, miHTT-155, for total-knockdown, and miSNP67T-155, for allele-selective knockdown. Both miRNA expression cassettes included a sequence that encodes green fluorescent protein (GFP) to evaluate transduction efficiency. [1] **Figure 11** presents the design of a proof-of-concept study to test miRNA performance in a HD rat model.

The engineered mmu-miHTT-155 constructs are suitable for initial experiments examining efficacy and distribution of the vector. However, the possibility of an immune reaction to GFP and off-target silencing makes them unsuitable for clinical trials in humans. For these reasons, a new miHTT precursor was presented for total knockdown. For allele-selective knockdown, an optimized construct designed to preferentially bind to HTT messenger RNA carrying SNP associated at high frequencies with HD was introduced. The resultant constructs are named miHTT-451 and miSNP50T-451.

E. Delivery Strategies for the Vector

Efficient delivery of the AAV5 vector is crucial for therapeutic success. The primary method of administration involves direct intracranial injections into the striatum, a region of the brain severely affected by Huntington's disease. This approach ensures localized delivery of the vector, maximizing its bioavailability and reducing off-target effects [3]. In addition, studies have explored less invasive delivery

methods, such as intrathecal injections, which could provide a broader distribution throughout the central nervous system but may require higher doses to achieve similar therapeutic outcomes [1].

In this study, AAV5-miHTT is injected directly into the striatum of the mice (13), a critical site for HTT protein expression. **Figure 12** shows the experimental setup used for administering the vector into the brains of R6/2 mice, where bilateral injections at 5x increasing doses were performed at 4 weeks of age, followed by behavioral testing and clinical measurements at 28 weeks of age. Extra wild-type littermates and Q175 KI mice received vehicle injections and acted as control groups. [1]

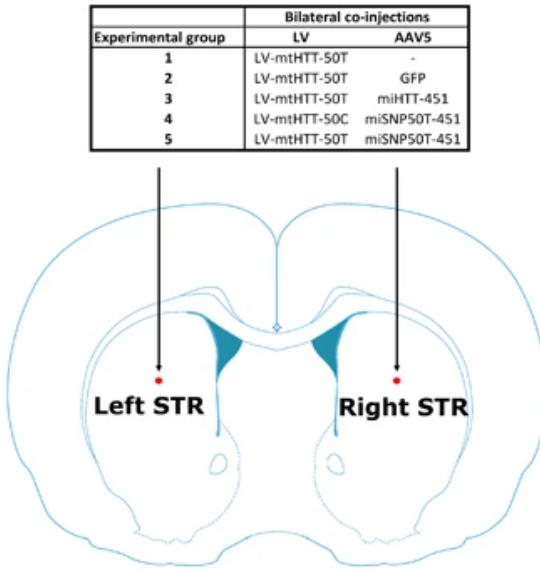


FIGURE 13. Bilateral co-injections in the striatum (STR) of rats with LV-mtHTT-50C or LV-mtHTT-50T, and AAV5-miHTT-451 or AAV5-miSNP50T-451 vectors. The experimental groups and injection sites are outlined. [1]

F. Targeted Distribution and Bioavailability

An important aspect of the vector's effectiveness is its distribution in the brain and its ability to specifically target regions affected by the disease. The AAV5 vector exhibits a strong affinity (tropism) for neural tissues, making it particularly suitable for treating Huntington's disease. Upon administration, the vector particles traverse the blood-brain

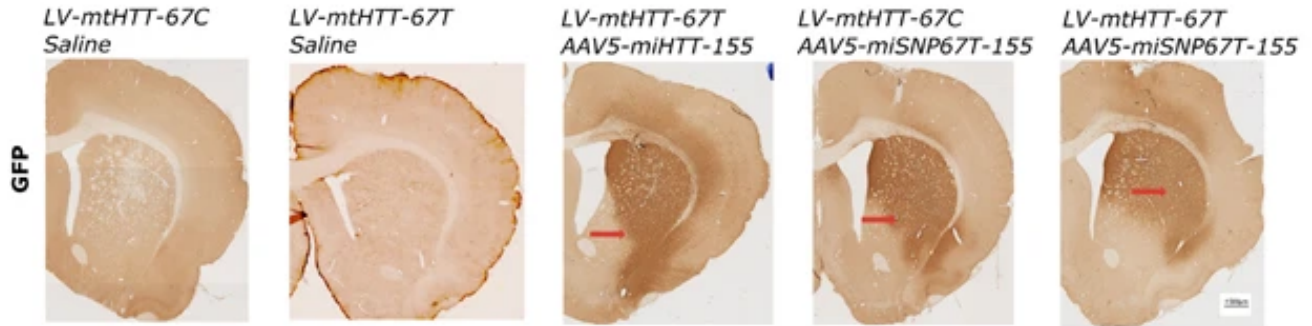


FIGURE 14. IHC with an anti-GFP antibody showing AAV5 distribution in the right hemisphere. GFP-positive areas are depicted by a red arrow. [1]

barrier and preferentially infect neurons in the striatum and cortex [3]. The high affinity of AAV5 for these brain regions is attributed to specific cell surface receptors that facilitate viral entry, enhancing the vector's targeting precision. The enhanced bioavailability of AAV5 in neural tissue supports its use in long-term gene therapy, as the vector can remain active and express the therapeutic gene for extended periods without triggering significant immune responses [1].

Figure 14 shows the distribution of the AAV5-miHTT-155 vector in the rat brain, visualized by immunohistochemistry (IHC) using an anti-GFP antibody, which allows the localization of areas where the vector is present. It represents the bioavailability of the vector in brain tissue and its ability to reach targeted regions such as the striatum.

Although the cortex was not the intended target area for the vector administration, varying levels of vector DNA were detected in the cortex of animals treated with AAV5-miHTT. Vector DNA and dose-dependent miHTT expression in the cortex remained consistent 12 months following a injection of AAV5-miHTT in Q175 mice. Since the miHTT transgene is present in the cortex, a brain region not directly targeted by AAV5-miHTT administration, this suggests that the AAV5 vector was effectively circulated throughout the brain via axonal transport. GFP expression in mouse brains that received AAV5-GFP with the same administration and dosage range similarly demonstrates a dose-dependent, extensive spread of the vector to both the striatum and cortex (**figure 15**). [3]

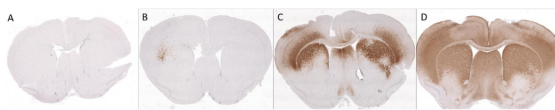


FIGURE 15. GFP staining of mouse brain after bilateral intra-striatal injection with formulation buffer (A) or AAV5-GFP at low (B), mid (C) or high (D) dose. [3]

G. Mechanisms of Therapeutic Action

The primary mechanism of action for the AAV5-miHTT vector involves RNA interference. Once inside the target cells, the AAV5 vector releases its miRNA payload into the cytoplasm, where it engages the RNA-induced silencing

complex (RISC). The miRNA directs the RISC to bind the HTT mRNA, leading to its degradation and preventing the synthesis of the toxic Huntingtin protein [3]. By lowering the levels of this protein, the treatment aims to alleviate neuronal stress and prevent cell death, potentially slowing disease progression and improving motor and cognitive functions in patients.

Figure 16 Figure 16 demonstrates a dose-dependent reduction in mutant Huntingtin levels in the striatum and cortex following a single injection of AAV5-miHTT. These findings highlight the vector's effectiveness in mitigating HTT protein toxicity, a critical factor in the progression of Huntington's disease. Immunohistochemical analyses confirm the vector's widespread distribution and prolonged bioactivity in targeted tissues.

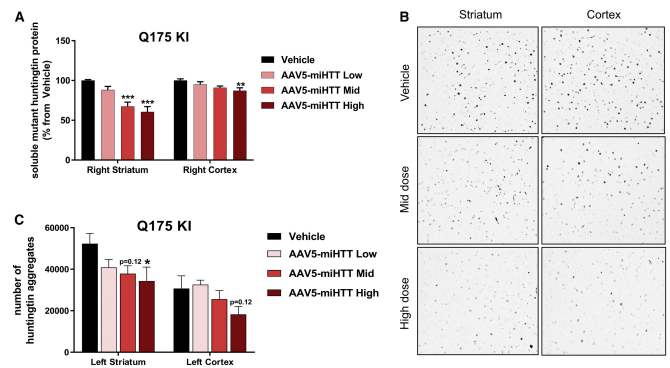


FIGURE 16. Reduction of mutant Huntingtin in the striatum and cortex of Q175 mice after treatment with AAV5-miHTT. Immunohistochemical analyses reveal a dose-dependent decrease in toxic aggregates, confirming the effectiveness of RNA interference in targeting mutant HTT proteins. [3]

H. Preclinical validation

Preclinical studies using mouse models of Huntington's disease, such as the Q175 and R6/2 strains, have demonstrated the efficacy of the AAV5-miHTT therapy. In Q175 mice, a single injection of AAV5-miHTT led to a sustained reduction in mutant HTT protein levels for up to 12 months post-injection, with no significant adverse effects observed [3].

In the more aggressive R6/2 mouse model, the therapy improved motor coordination and extended survival, suggesting a neuroprotective effect.

These findings, although preliminary, highlight the potential of AAV5-based therapies to provide a one-time, long-lasting treatment option for Huntington's disease.

Figure 17, which presents data on motor coordination improvements after treatment. Untreated R6/2 mice had a significantly lower latency to fall on the rotarod test compared to wild-type mice. Treatment with a high dose of AAV5-miHTT resulted in a 47% increase in latency, suggesting enhanced motor performance due to reduced neurotoxicity.

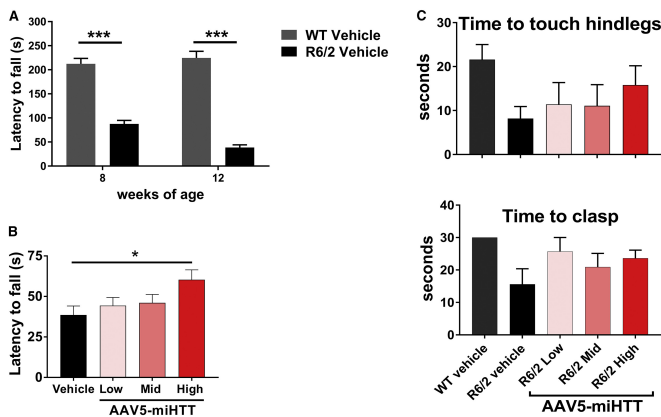


FIGURE 17. AAV5-miHTT Treatment Shows Motor Coordination Improvement in R6/2 Mice. [3]

Furthermore, the survival benefits of AAV5-miHTT were assessed using Kaplan-Meier analysis. **Figure 18** displays the survival curves for treated and untreated R6/2 mice. The results indicate a significant increase in median survival for both low- and high-dose treatment groups, highlighting the potential neuroprotective effect of the therapy.

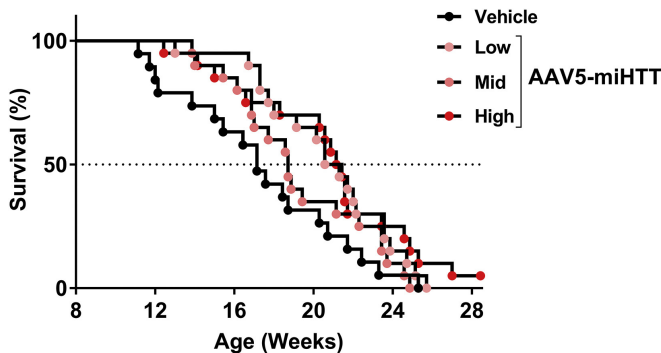


FIGURE 18. Kaplan-Meier Curve Showing Survival of Untreated and AAV5-miHTT-Treated R6/2 Mice. [3]

I. Limitations and Perspectives for Improvement

Despite its promise, the use of AAV5 vectors faces several challenges. One major concern is the immune response, as

patients may have pre-existing antibodies against AAV5, which could neutralize the vector before it reaches its target. To assess the immune reaction to AAV5 vectors that produce therapeutic miRNA precursors in the brain, levels of immunoreactivity were observed for each miRNA construct and compared against saline control. The results suggested no noticeable activation of immune response from microglia and astrocytes (**figure 19**). [1]

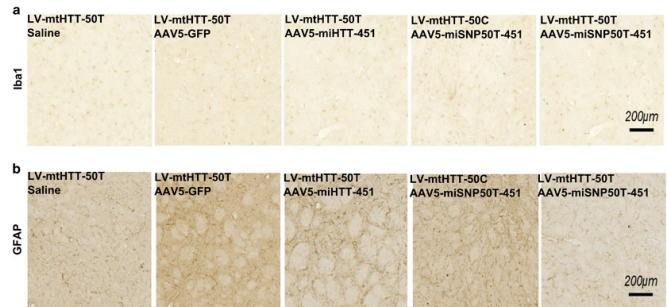


FIGURE 19. No apparent activation of immune response following AAV5-miHTT-451 and AAV5-miSNP50T-451 injections in HD rats. (a) IHC against Iba1 to show microglial activity. (b) IHC against GFAP to show the astrocyte activity. [3]

Furthermore, allele-selective knock-down still faces several challenges. Even though most patients with Huntington's disease (HD) having one normal copy and one mutant copy of the HTT gene, methods of distinction of the two alleles can vary from patient to patient, making it impossible to completely address the HD population with only one product. Total-knockdown allows to provide a treatment for all HD patients independent from their genotype. In spite of the concern about normal HTT production, preclinical research involving HD animal models and non-human primates suggests that a therapeutic window exists where complete HTT knock-down is both safe and advantageous for HD patients. [1]

Additionally, the limited packaging capacity of AAV5 (4.7 kb) restricts the size of the genetic material that can be delivered. Future research is needed to enhance vector design, including the development of capsid variants with reduced immunogenicity and improved targeting capabilities [3].

In summary, the AAV5 vector represents a significant advancement in the field of gene therapy for Huntington's disease. Its ability to specifically target neural tissues and deliver a potent miRNA payload offers a novel approach to mitigating the toxic effects of mutant HTT protein (**figure 20**). Future improvements may involve engineering capsid variants with lower immunogenicity and optimizing the RNA interference sequence for enhanced specificity and efficiency. Ongoing research and clinical trials will be critical in optimizing this strategy and addressing its current limitations, paving the way for potential breakthroughs in the treatment of neurodegenerative disorders.

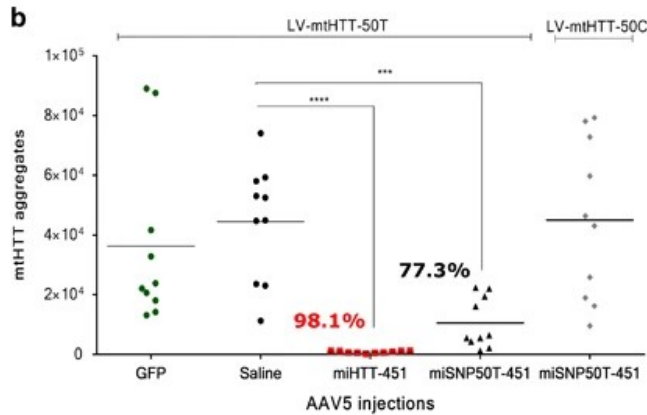


FIGURE 20. Quantification of anti-HTT staining shows a reduction of mutant HTT aggregates induced by the AAV5-miHTT-451 and AAV5-miSNP50T-451. The reduction (%) of mutant HTT aggregates is relative to the saline control. [1]

VI. Conclusion

Huntington's Disease exemplifies the immense challenges posed by neurodegenerative disorders, from their symptom diversity and lack of treatment options that may require the use of multiple medications to alleviate symptoms. The blood-brain barrier (BBB) is a significant obstacle for drug delivery, requiring innovative transportation methods to ensure effective treatment. The progress in vectorization techniques, particularly the use of adeno-associated viral vectors, offers a way to overcome some of these challenges. Vectors allow for the precise delivery of therapeutic agents in vehicles compatible with the human nervous system, bypassing the biological barriers and delivering genetic material directly to affected cells. In the study of Huntington's Disease, these methods have the potential to reduce mutant protein levels and slowing the course of the disease by directly targeting the mutant gene responsible for the disorder, as demonstrated in preclinical studies conducted in mouse models. Although obstacles like immune response and delivery efficiency persist, ongoing research continues to refine these techniques. The advancements discussed today reflect a major step toward innovative, long-term solutions for Huntington's and similar disorders.

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